

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech Theory Examination (Mar-Apr 2018)

CL216.01 Mathematics for Civil Engineers

Date: 26.03.2018, Monday

Time: 01.30 p.m. To 04.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper contains two sections.
2. Each section has three questions.
3. Section I and II must be attempted in separate answer sheets.
4. Each section carries 35 marks.
5. Figures to the right indicate marks.
6. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 Multiple Choice Questions

[07]

- 1) Inverse Laplace transform of $\frac{12}{s^2 - 9}$ is
 (a) $3\sinh 4t$ (b) $4\sinh 3t$ (c) $4\cosh 3t$ (d) $3\cosh 4t$
- 2) A biquadratic equation has _____ roots.
 (a) 3 (b) 4 (c) 1 (d) 2
- 3) The expression $\text{curl}(\text{grad } f)$, where f is a scalar function, is
 (a) equal to $\nabla^2 f$ (b) equal to $\text{div}(\text{grad } f)$ (c) a scalar of zero magnitude
 (d) a vector of zero magnitude
- 4) Laplace transform of e^{2t+3} is
 (a) $\frac{e^3}{s-2}$ ($s > 2$) (b) $\frac{e^2}{s-3}$ (c) $\frac{1}{s-\log 2}$ (d) $\frac{1}{s-2}$
- 5) $x^3 + 10 = 0$ has
 (a) One negative root (b) one positive root (c) 3 complex roots (d) one positive and two complex roots
- 6) A vector $\vec{F}$ is solenoidal if
 (a) $\text{div}\vec{F} = 0$ (b) $\text{curl}\vec{F} = 0$ (c) $\text{div}\vec{F} \neq 0$ (d) $\text{curl}\vec{F} \neq 0$
- 7) $L^{-1}\left(\frac{1}{s^n}\right)$ is possible when n is
 (a) Zero (b) negative integer (c) positive integer (d) negative rational

Q - 2 Attempt any **four** out of **six**

[16]

- 1) Solve the differential equation $y'' + 5y' + 6y = 0$ and $y(0) = 0$, $y'(0) = 1$ using Laplace transform.
- 2) Find the solution of the equation $x^3 - 18x - 35 = 0$ by Cardan's method.

- 3) A particle moves along the curve $x = 2t^2, y = t^2 - 4t, z = 3t - 5$, where t is the time. Find the components of its velocity and acceleration at time $t=1$ in the direction $\hat{i} + 2\hat{j} + 2\hat{k}$.
- 4) Find the Laplace transforms of :
- (i) $e^{-t} \cos t$ (ii) $\sin t \cos t$.
- 5) Find the solution of the equation $x^4 + 12x - 5 = 0$ by Ferrari's method.
- 6) Show that $\vec{F} = (y^2 + 2xz^2)\hat{i} + (2xy - z)\hat{j} + (2x^2z - y + 2z)\hat{k}$ is irrotational and hence find its scalar potential.

Q - 3 Attempt any **two** out of **three**

[06]

- (a)
- 1) Using the convolution theorem, find $L^{-1} \left\{ \frac{1}{s^2(s-1)} \right\}$.
- 2) Solve the equation $x^3 - 3x^2 + 1 = 0$ by Cardan's method.
- 3) Prove that $r^n \vec{r}$ is irrotational.
- Q-3(b) Attempt any **two** out of **three**.
- [06]
- 1) Find the inverse Laplace transform of the function $F(s) = \frac{1}{s^2 + s}$.
- 2) Solve the equation $x^4 + 2x^3 - 7x^2 - 8x + 12 = 0$ by Ferrari Method.
- 3) Find the directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at the point $P(2, 1, 3)$ in the direction of the vector $\vec{a} = \hat{i} - 2\hat{k}$.

SECTION - II

Q - 4 Multiple Choice Questions

[07]

- 1) A necessary and sufficient condition that line integral $\oint_C \vec{F} \cdot d\vec{r} = 0$ for every closed curve C is that
- (a) $\text{div } \vec{f} = 0$ (b) $\text{curl } \vec{f} = 0$ (c) $\text{div } \vec{f} \neq 0$ (d) $\text{curl } \vec{f} \neq 0$
- 2) If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then $\text{curl } \vec{r}$ equal to
- (a) 3 (b) r (c) 0 (d) $-r$
- 3) If a box contains 6 red and 4 blue balls, the probability of drawing first the red and then blue ball is $P = \dots\dots\dots$
- (a) $1/210$ (b) $1/120$ (c) $1/201$ (d) $1/102$
- 4) The Gauss divergence theorem relates certain surface integrals to
- a) Line integral (b) Volume integral (c) Surface integral (d) None of above
- 5) A coin is tossed 3 times. The probability of obtaining two heads will be
- (a) $3/8$ (b) $1/2$ (c) 1 (d) 2

- 6) E equal to
 (a) $1 + \Delta$ (b) $\Delta \nabla$ (c) $\Delta + \nabla$ (d) $\Delta - \nabla$
- 7) The order of the difference equation $y_{n+3} - 3y_{n+1} + 2y_n = 0$ is
 (a) 1 (b) 2 (c) 3 (d) 0

Q -5 Attempt any **four** out of **six**

[16]

- 1) Evaluate $\int_C y^2 dx + x dy$, where C is the arc of parabola $x = 4 - y^2$ between (-5, -3) and (0, 2).
- 2) 100 electric bulbs are found to be defective in a lot of 500 bulbs. Find the probability that at the most 3 bulbs are defective in a box of 100 bulbs. [$e^{-2} = 0.1353$].
- 3) Compute seven decimal value of the Bessel function $J_0(x)$ for $x = 1.72$ from the four values in the following table, using Newton's forward formula

X:	1.7	1.8	1.9	2.0
$J_0(x)$:	0.397 9849	0.339 9864	0.281 81816	0.223 8908

- 4) Use Green's theorem, to evaluate $\oint_C xy dx + x^2 y^3 dy$, where C is the triangle with vertices (0,0), (1,0) and (1,2).
- 5) An unbiased coin is tossed 6 times. Find the probability of getting
 (i) exactly 4 heads (ii) at least 4 heads .
- 6) Using Lagrange's formula find a polynomial of degree 3 which fits into the data below:

X:	-1	0	1	3
F:	2	1	0	-1

Q -6 Attempt any **two** out of **three**

[06]

(a)

- 1) Using Divergence theorem, evaluate the surface integral $\iint_S (yz dy dz + xz dz dx + xy dy dx)$ where $S : x^2 + y^2 + z^2 = 4$.
- 2) Find the binomial distribution for $n=4$ and $p=0.3$.
- 3) If Δ, ∇, δ denote forward, backward and central difference operators and E, μ are respectively the shift and averaging operators, then show that
 (i) $\nabla = 1 - E^{-1}$ (ii) $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$ (iii) $\mu = \frac{1}{2} \left(E^{\frac{1}{2}} + E^{-\frac{1}{2}} \right)$

Q -6 Attempt any **one** out of **two**

[06]

(b)

- 1) Verify Stoke's theorem for $\vec{F} = (x^2 - y^2)\hat{i} + 2xy\hat{j}$ in the rectangular region bounded by $x = 0, x = a, y = 0, y = b$
- 2) A card is drawn at random from a pack of 52 cards. What is the probability that the card is a spade or a king?